

Project Development Phase

Preprocessing Steps And Business Questions With Visualization

Date	7 july 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Data preprocessing steps

Step No	Preprocessing Step	Description
1	Import Dataset	Import the electricity consumption dataset (CSV/Excel) into Tableau.
2	Inspect Data	Verify column names, data types, and overall dataset structure.
3	Remove Duplicates	Identify and remove duplicate records to ensure data accuracy.
4	Handle Missing Values	Replace, remove, or fill null values where appropriate.
5	Correct Data Types	Convert Date, Numeric, and Categorical fields into appropriate formats.
6	Rename Columns	Use meaningful column names for easier analysis.
7	Filter Invalid Record	Remove incomplete or invalid entries from the dataset.
8	Create Calculated Fields	Create KPIs such as Total Consumption, Average Consumption, Growth Rate, etc.
9	Sort & Aggregate Data	Group data by Year, Month, Region, or Sector for analysis.
10	Validate Dataset	Verify data consistency before creating visualizations.

Business Questions with Recommended Visualizations

S.No	Business Question	Recommended Visualization
1	What is the total electricity consumption over time?	Line Chart
2	Which region has the highest electricity consumption?	Horizontal Bar Chart
3	Which sector consumes the most electricity?	Pie Chart or Tree map

4	How does electricity consumption vary month by month?	Line Chart
5	What are the yearly electricity consumption trends?	Area Chart or Line Chart
6	Which months experience peak electricity usage?	Heat Map
7	How does electricity consumption differ across regions?	Filled Map
8	Which sector contributes the highest percentage of electricity consumption?	Donut Chart or Pie Chart
9	What are the top 10 electricity-consuming regions?	Horizontal Bar Chart
10	What is the average electricity consumption by sector?	Bar Chart